

CY 501: Organic Reactions & Mechanism

Course Code	CY 501
Course Name	Organic Reactions & Mechanism
L-T-P-C	3-1-0-4
Prerequisites	B.Sc. (with Chemistry) or Teachers consent Intended for: UG/PG
Distribution	Core Semester: Odd/Even
Approval	8th Senate (Oct 2015)

Course Contents

- **Nucleophilic Substitution Reaction**

Aliphatic Nucleophilic Substitution Reaction: The SN₂, SN₁, borderline of SN₁ and SN₂ mechanisms involving ion pairs and single electron transfer (SET) mechanisms. Relationship between stereochemistry and reaction mechanism. Neighbouring Group Participation, anchimeric assistance. Classical and nonclassical carbocations, phenonium ions, norbornyl system, common carbocation rearrangements. The SN_i mechanism. Nucleophilic substitution at an allylic, aliphatic trigonal and a vinyl carbon. Reactivity effects of substrate structure, attacking nucleophile, leaving group and reaction medium. [14 Lectures]

- **Aromatic Nucleophile Substitution**

The S_NAr, SN₁, benzyne and SRN₁ mechanisms. Reactivity; effect of substrate structure, leaving group and attacking nucleophile.

- **Electrophilic Substitution Reaction**

Aliphatic electrophilic substitution reaction: SE₁, SE₂ and SE_i. The SE₁ mechanism, electrophilic substitution accompanied by double bond shifts. Effect of substrates, leaving group and the solvent polarity on the reactivity.

- **Aromatic electrophilic substitution reaction**

The arenium ion mechanism, orientation and reactivity, energy profile diagrams. The ortho/para ratio, ipso attack, orientation in other ring systems. Quantitative treatment of reactivity in substrates and electrophiles. [10 Lectures]

- **Elimination Reactions**

The E₂, E₁ and E₁cB mechanisms and their spectrum. Regiochemistry of elimination reactions and stereochemistry of E₂ elimination reactions. Reactivity; effects of substrate structures, attacking base, the leaving group and the medium. Concerted pyrolytic eliminations. [5 Lectures]

- **Free Radical Reactions**

Types of free radical reactions, free radical substitution mechanism, mechanism at an aromatic substrate, neighbouring group assistance. Reactivity for aliphatic and aromatic substrates at a bridgehead. Reactivity in the attacking radicals. The effect of solvents on reactivity. Addition to Carbon-Carbon Multiple Bonds: Free radical mediated cyclization reactions Baldwin rules. [6 Lectures]

- **Mechanistic and stereochemical aspects of other reactions**

Addition reactions involving electrophiles, nucleophiles and free radicals, regio- and chemo- selectivity, orientation and reactivity. Addition to cyclopropane ring. Hydroboration. Addition to Carbon-Hetero Multiple Bonds: Mechanism of metal hydride reduction of saturated and unsaturated carbonyl compounds, acids, esters and nitriles. Addition of Grignard reagents, organozinc and organolithium reagents to carbonyl and unsaturated carbonyl compounds. [7 Lectures]

Text Books

1. F. A. Carey and R. A. Sundberg, **Advanced Organic Chemistry, Part A: Structure and Mechanisms**, 5th edition, Springer, New York, 2007
2. Reinhard Bruckner, **Advanced Organic Chemistry Reaction mechanism**, Academic Press, 2002.
3. Jonathan Clayden, Nick Greeves, Stuart Warren, **Organic Chemistry**, Oxford University Press, 2001
4. J. March, **Advanced Organic Chemistry**, John Wiley & Sons, 1992
5. S. H. Pine, **Organic Chemistry**, McGraw Hill, 1987.

References

1. H. O. House, **Modern Synthetic Reactions**, W.A. Benjamin, 1972.
2. A. Jacobs, **Understanding Organic Reaction Mechanism**, Cambridge 1998.
3. J. M. Hornback, **Organic Chemistry**, Books Coley, 1998.
4. P.Y. Bruice, **Organic Chemistry**, Prentice Hall, 1998.
5. P.S. Kalsi, **Organic Reaction and their Mechanism**, New Age, 1996.
6. Goutam Brahmachari, **Organic Chemistry Through Solved problems**, Narosa Publishing House, 2014.